

Jacob Yankee
COSC 461
Maniccam

A. Number of epochs

Convolutional layers: 2. Number of convolutions: 50. Size of convolutions: (3, 3). Pool size: (2, 2). Pool type: Max.

Epochs	Accuracy	Time (seconds)
2	0.8928	158
4	0.9048	291
6	0.9100	417
8	0.9104	595
10	0.9142	723

Summary: Increasing the number of epochs increases the accuracy ever so slightly at the cost of greatly increased runtimes. As increasing the number of epochs increases the complexity of the neural network, the runtime increases as well.

B. Number of convolutional layers

Epochs: 2. Number of convolutions: 50. Size of convolutions: (3, 3). Pool size: (2, 2). Pool type: Max.

Number of layers	Accuracy	Time (seconds)
1	0.8826	81
2	0.8928	158
3	0.8824	178
4	0.8868	209
5	0.8730	217

Summary: Increasing the number of convolutional layers slightly increases accuracy going from 1 to the control value of 2, but after that it starts to decrease in addition to increasing runtime. This can be seen as a case of overfitting as adding more layers increases the complexity too much.

C. Number of convolutions

Epochs: 2. Convolutional layers: 2. Size of convolutions: (3, 3). Pool size: (2, 2). Pool type: Max.

Number of convolutions	Accuracy	Time (seconds)
25	0.8738	85
50	0.8928	158
75	0.8831	247
100	0.8963	322
150	0.8876	585

Summary: Increasing the number of convolutions mainly increases the runtime, however the accuracy fluctuates ever so slightly. This can be seen as overfitting as the complexity of the algorithm causes the accuracy to be inconsistent as convolutions increase. The runtime increases happen for the same reason as a more complex algorithm takes longer to run.

D. Size of convolutions

Epochs: 2. Convolutional layers: 2. Number of convolutions: 50. Pool size: (2, 2). Pool type: Max.

Size of convolutions	Accuracy	Time (seconds)
2x2	0.8849	120
3x3	0.8928	158
4x4	0.8806	155
5x5	0.8861	186
6x6	0.8826	171

Summary: The accuracy peaks at the control value of 3x3 and decreases as the size increases with slight increases to runtime. The convolution size of 3x3 is already more than enough for the dataset, so making it larger is unnecessary.

E. Pool size

Epochs: 2. Convolutional layers: 2. Number of convolutions: 50. Size of convolutions: (3, 3). Pool type: Max.

Pool size	Accuracy	Time (seconds)
1x1	0.9083	525
2x2	0.8928	158
3x3	0.8791	86
4x4	0.8638	75
5x5	N/A	N/A

Summary: Increasing the size of the pool type decreases both the accuracy and the runtime with the exception of the 5x5 pool which cannot run due to being larger than the feature map. Since increasing the pool size decreases the size of the feature map, this allows the runtime to improve while sacrificing very little accuracy.

F. Pool type

Epochs: 2. Convolutional layers: 2. Number of convolutions: 50. Size of convolutions: (3, 3). Pool size: (2, 2).

Pool type	Accuracy	Time (seconds)
Max	0.8928	158
Avg	0.8782	128

Summary: Using average pooling instead of max pooling slightly improves runtime while slightly reducing the accuracy. Since max pooling keeps track of finer details, having a higher accuracy and runtime makes sense compared to average pooling which only tracks average values, resulting in a lower runtime.